

Backend Engineer Take-Home Assessment (2–3 Hours)

Overview

Build a simplified backend service that imports payout transactions from CSV, reconciles them against existing orders, and exposes reconciliation results via an API.

Technical Requirements

Python, Django, Django REST Framework, SQLite or PostgreSQL, Git.

Part 1 – Import API

Create POST `/api/payouts/upload` to import payout CSV (provider, order_number, amount, currency) and return the number of imported records.

Part 2 – Reconciliation

Implement statuses: Matched, Missing Order, Amount Mismatch. (Optional: Currency Mismatch.) Store reconciliation results.

Part 3 – Results API

Create GET `/api/reconciliation` returning order_number and reconciliation status.

Design Discussion

In the README, explain how you'd scale to 500,000 records, which AWS services you'd use, and production improvements.

AI Usage

Briefly describe any AI coding tools used and how you validated generated code.

Submission

Submit GitHub repository, README, tests, sample CSV, and setup instructions.

Sample orders.csv

```
order_number,total_amount,currency
100001,125.00,USD
100002,89.99,USD
100003,240.50,USD
100004,45.00,USD
100005,175.25,USD
100006,60.00,USD
```

100007, 320.00, USD
100008, 99.95, USD
100009, 18.50, USD
100010, 500.00, USD

Sample payouts.csv

provider,order_number,amount,currency
Stripe,100001,125.00,USD
Stripe,100002,89.99,USD
Stripe,100003,250.50,USD
Stripe,100004,45.00,USD
Stripe,100006,60.00,USD
Stripe,100008,99.95,USD
Stripe,100009,18.50,CAD
Stripe,100011,75.00,USD
Stripe,100012,120.00,USD